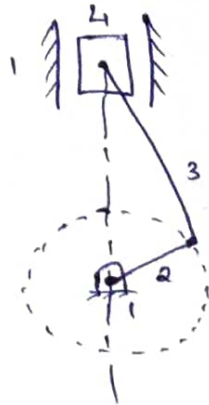


19-8-24.

Links form linkage.

When any one link is fixed, then it is called a mechanism.

Connecting rod is also called coupler.



4-bar linkage or,
4-bar mechanism
or, slider-crank mechanism.

Moving rigid bodies can be considered as links.

Joints are also called kinematic pairs.

- Lower pair - have surface or area of contact b/w 2 links.
- Higher pair - Point or line of contact.
Eg - gears, cams...
- Wrapping pair - Eg: belt & pulley.

Naming by no. of connections:

⇒ Binary 

⇒ Ternary 

⇒ Quaternary 

Degrees of freedom (DOF)

A Rigid body in space has 6° of freedom.

3 translational & 3 rotational.

Rigid body in a plane should have max 3 DOF.

Spatial & planar mechanism.
↓ ↓
in space in a plane.

DOF of a planar mechanism $\Rightarrow 3(n-1)$

DOF of fixed link = 0.

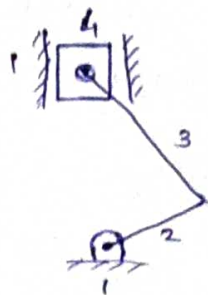
$n \Rightarrow$ no. of links.

For each joint, 2 DOF is reduced.

if j = no. of joint (lower pair).

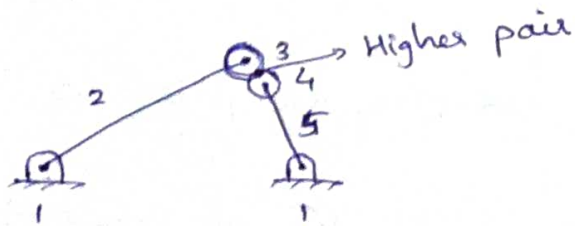
2 DOF reduces.

Sliding or prismatic pair. } lower pair.
Revolution or turning pair }



3R - 1P mechanisms.

first 4 & 5 were considered as 1.



$n = 4$, $j = 4$, h = no. of higher pair.

$$\therefore \text{DOF} = 3(n-1) = 3(5-1) = 12 - 8 = \underline{4}$$

For each higher pair, 1 DOF reduces.

$$\boxed{\text{DOF} = 3(n-1) - 2j - h - F_x}$$

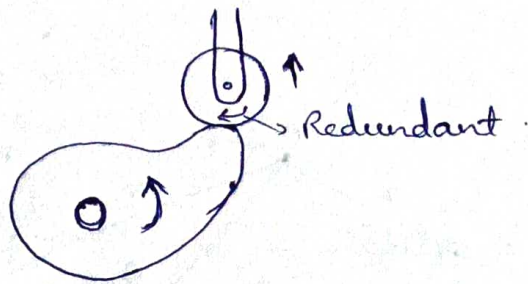
$\Rightarrow$ Grubler's criteria
or,

Kutzbach criteria.

F_x = redundant force.

Redundant DOF $\Rightarrow$ unwanted DOF

Eg: cam-follower.



Instantaneous centre method.

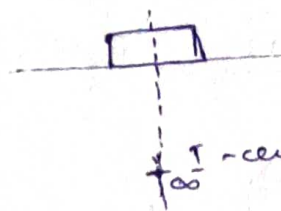
Asshold Kennedy Theorem.

If three bodies are in relative motion, in one plane, then their I-centres are collinear. i.e., lies in the same line.

Revolute pair.

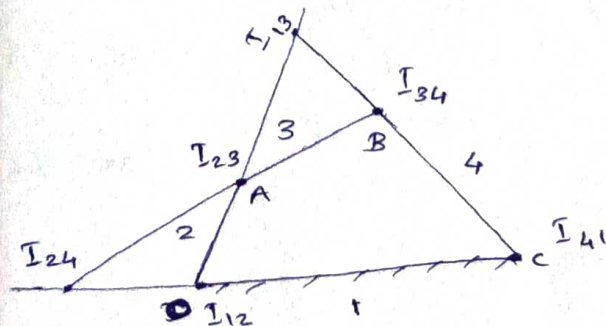


sliding pair

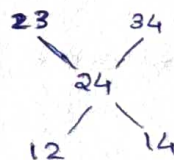
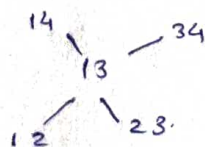
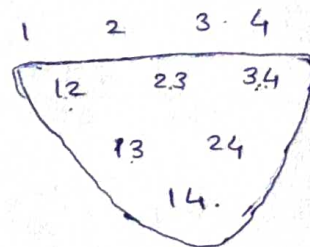


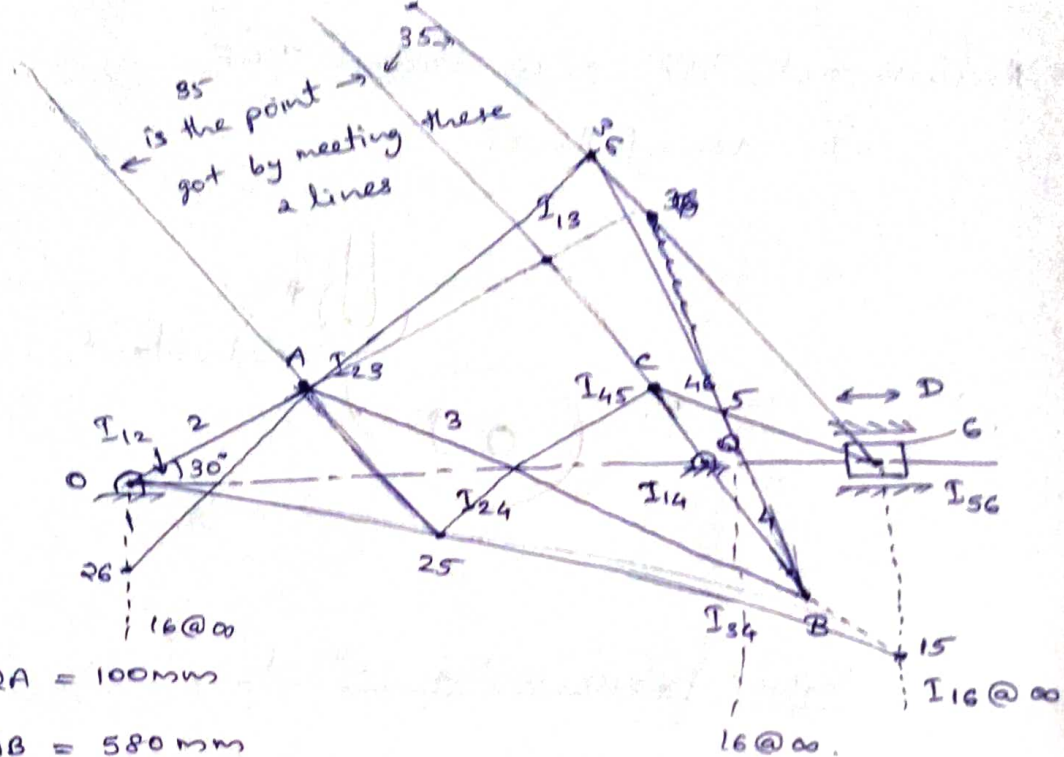
$$I\text{-centre} \Rightarrow \frac{n(n-1)}{2}$$

where, $n = \text{no. of links.}$



$$\therefore \frac{4 \times 3}{2} = \underline{6} \text{ links}$$





$$QA = 100 \text{ mm}$$

$$AB = 580 \text{ mm}$$

$$BC = 300 \text{ mm}$$

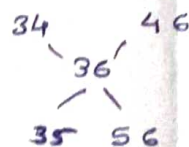
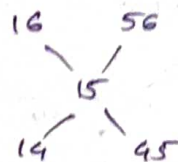
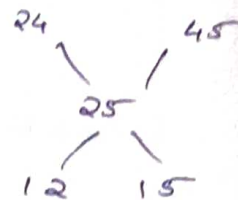
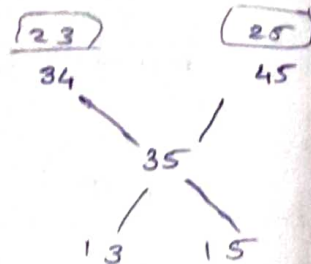
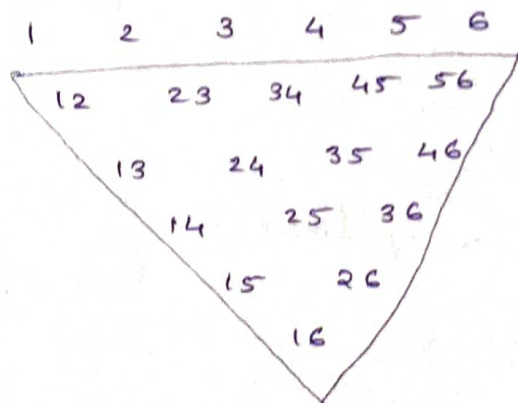
$$QC = 100 \text{ mm}$$

$$CD = 350$$

OA rotates 150 rpm clockwise.

Ans.

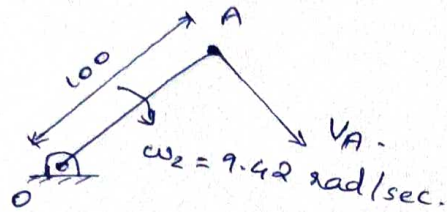
$$\frac{n(n-1)}{2} = \frac{6 \times 5}{2} = \underline{\underline{15}}$$



$$\omega_2 = \frac{2\pi N}{60} = 3\pi$$

$$\omega_2 = \underline{\underline{9.42 \text{ rad/sec.}}}$$

$$V_A = OA \times \omega_2 = 0.1 \times 9.42 = 0.94 \text{ m/s.}$$



$$V_A = V_{23} = \underline{\underline{0.94 \text{ m/s}}}$$

$$V_B = V_A \left(\frac{13-34}{13-23} \right) \quad (34)$$

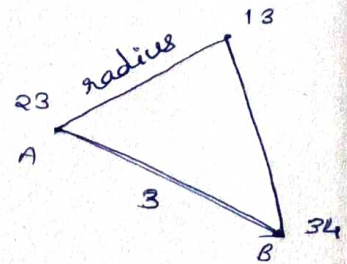
$$\frac{V_A (23)}{13-23} = \frac{V_B}{13-34}$$

$$V_B = V_A \left[\frac{13-34}{13-23} \right]$$

$$V_C = V_B \left[\frac{14-45}{14-34} \right]$$

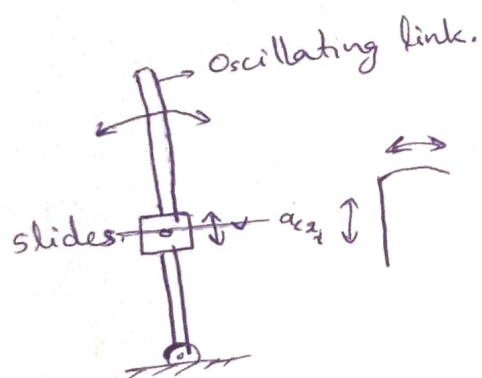
$$V_D = V_C \left[\frac{15-56}{15-45} \right]$$

similar problem assignment.



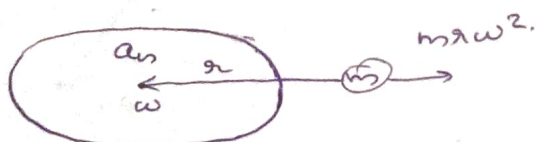
6-11-24.

Coriolis component of acceleration.



Coriolis component;

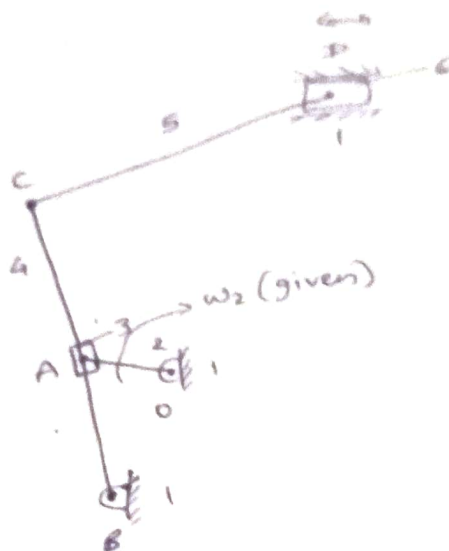
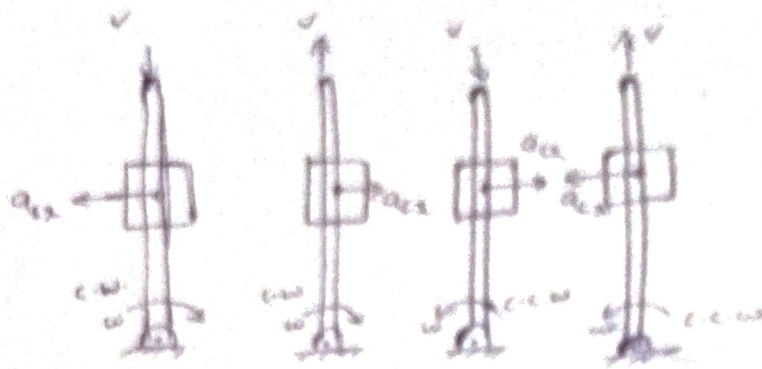
$$a_{cy} = 2v\omega$$



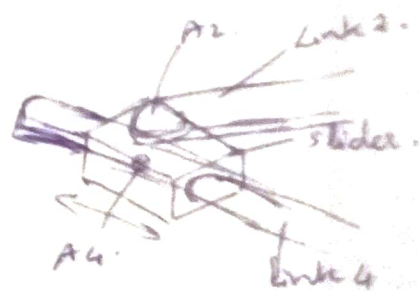
$$F \propto a \Rightarrow F = m a = m(r\omega^2)$$

$$\frac{v^2}{r} = a_n = r\omega^2$$

$$v = r\omega, \quad \omega = \frac{v}{r}, \quad a_n = r \cdot \frac{v^2}{r^2} = \frac{v^2}{r}$$

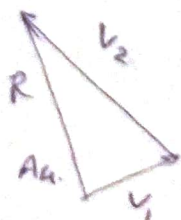
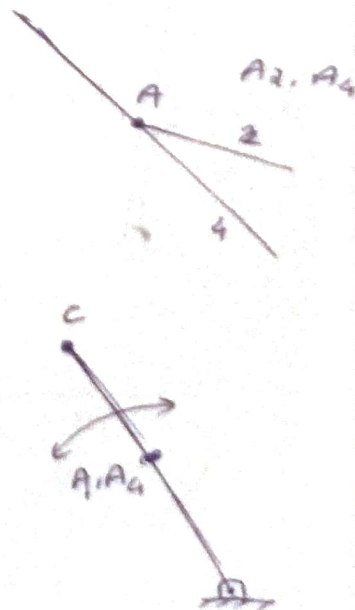
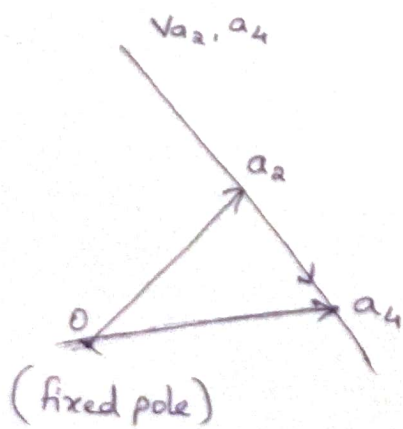


All dimensions are given, w_2 - given.



velocity polygon.

$$V_A = \omega_2 \times OA = OA \times \omega_2 \quad (\text{given})$$



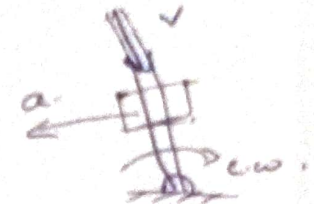
Acceleration Polygon.

$$A_{A_2} = A_{A_2t} + A_{A_2n}$$

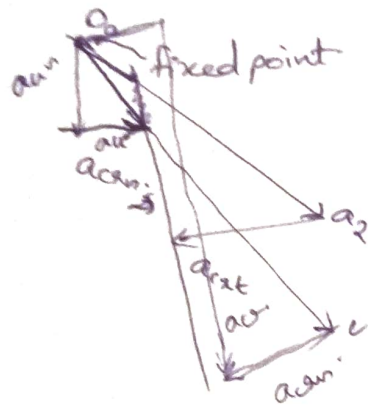
$$A_{A_2n} = \alpha_2 \times OA + \frac{V_0^2}{OA}$$

$$A_{A_2n} = \frac{V_{A_2}^2}{OA} = m/s^2$$

$$a_{ce} = 2V\omega_4$$



$$V_4 = r\omega_4$$



$$A_{A_4} = A_{A_2} + A_{A_4n}$$